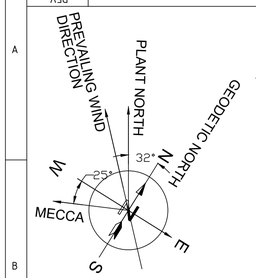
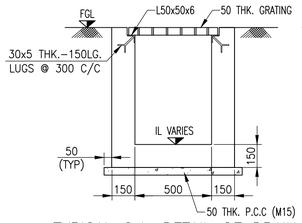
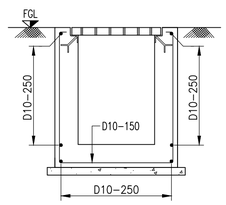
DRAWING NUMBER: MD-216-49EE-EG-CV-GAD-1585 REV D01





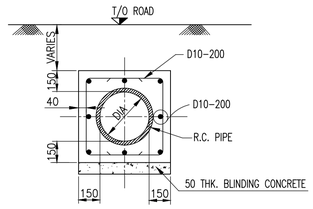
TYPICAL G.A. DETAIL OF DRAIN

SCALE 1:20



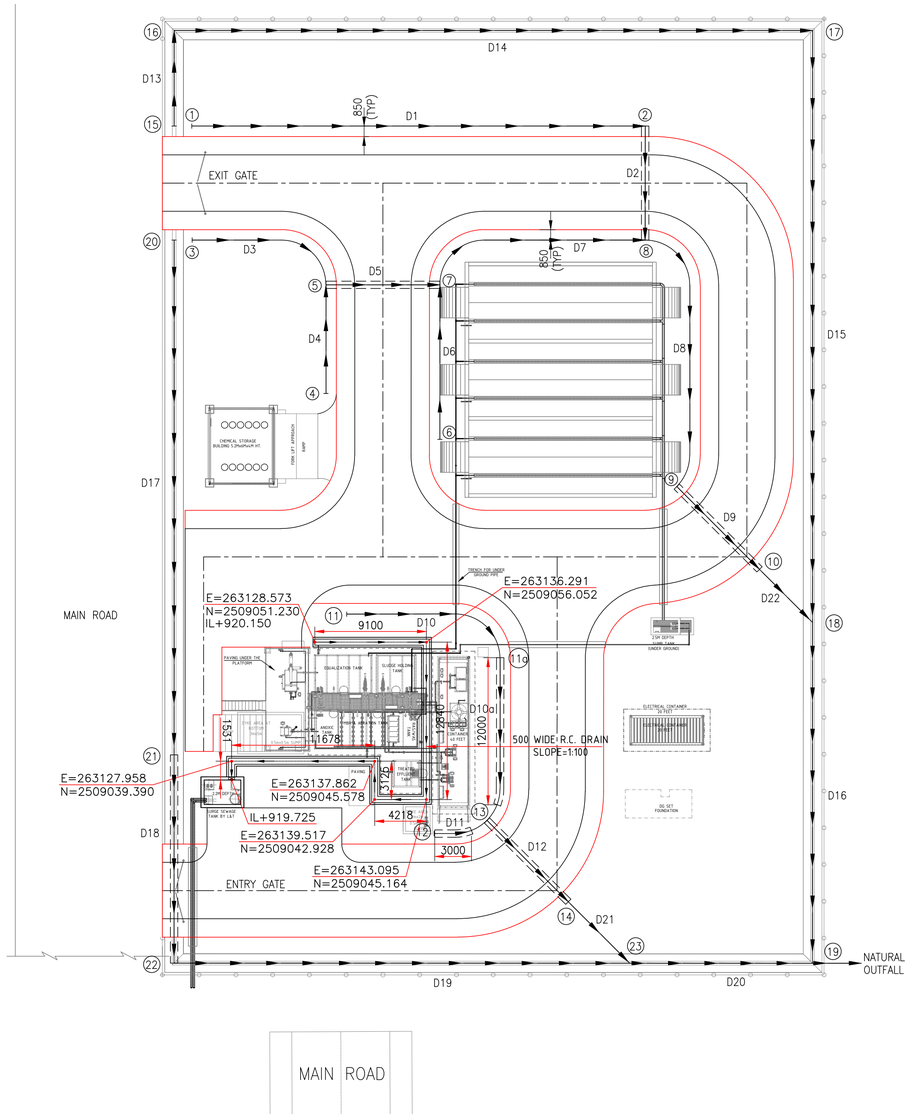
R.C. DETAIL OF DRAIN

SCALE 1:20



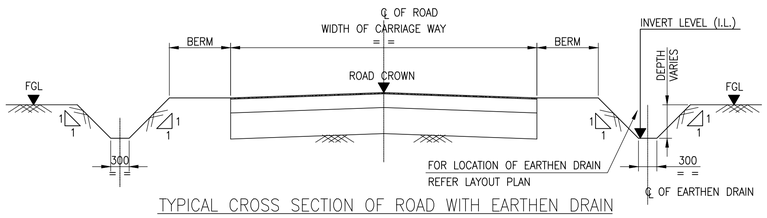
TYPICAL CROSS SECTION OF R.C. PIPE UNDER ROAD

SCALE 1:20



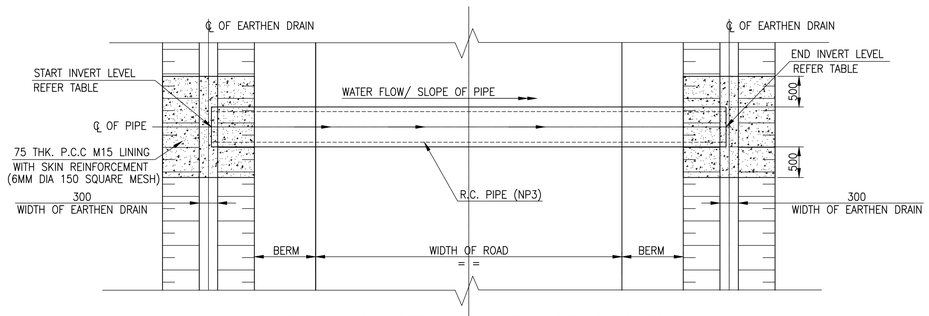
SCHEDULE OF STORM WATER DRAINAGE SYSTEM

SL. NO.	NODE START	NODE END	DRAIN MKD.	LENGTH (M)	CL. OF CO-ORDINATE (M) START NODE	CL. OF CO-ORDINATE (M) END NODE	INVERT LEVELS (M) START NODE	INVERT LEVELS (M) END NODE	SLOPE	DRAIN TYPE	SHAPE	REMARKS
1	1	2	D1	37.0	E=263097.740 N=2509081.616	E=263129.117 N=2509101.223	920.150	920.027	1:300	EARTHEN TRAPEZOIDAL	300	
2	2	8	D2	9.3	E=263129.117 N=2509101.223	E=263134.045 N=2509093.336	920.027	919.980	1:200	R.C. PIPE DRAIN	ø300	
3	3	5	D3	13.0	E=263102.695 N=2509073.746	E=263113.887 N=2509076.435	920.100	920.048	1:250	EARTHEN TRAPEZOIDAL	300	
4	4	5	D4	8.865	E=263118.584 N=2509068.917	E=263113.887 N=2509076.435	920.100	920.055	1:200	EARTHEN TRAPEZOIDAL	300	
5	5	7	D5	9.3	E=263113.887 N=2509076.435	E=263121.774 N=2509081.363	920.048	920.017	1:300	R.C. PIPE DRAIN	ø300	
6	6	7	D6	12.6	E=263128.456 N=2509070.668	E=263121.774 N=2509081.363	920.100	920.016	1:150	EARTHEN TRAPEZOIDAL	300	
7	7	8	D7	18.84	E=263121.774 N=2509081.363	E=263134.045 N=2509093.336	920.016	919.985	1:600	EARTHEN TRAPEZOIDAL	300	
8	8	9	D8	22.5	E=263134.045 N=2509093.336	E=263146.897 N=2509077.642	919.980	919.935	1:500	EARTHEN TRAPEZOIDAL	300	
9	9	10	D9	9.55	E=263146.897 N=2509077.642	E=263156.200 N=2509075.494	919.935	919.897	1:250	R.C. PIPE DRAIN	ø300	
10	11	11a	D10	14.52	E=263129.563 N=2509054.545	E=263142.067 N=2509058.169	920.150	920.102	1:300	EARTHEN TRAPEZOIDAL	300	
11	11a	13	D10a	13.5	E=263142.067 N=2509058.169	E=263148.194 N=2509046.334	920.102	920.057	1:300	R.C. PIPE DRAIN	ø300	
12	12	13	D11	4.6	E=263145.100 N=2509043.141	E=263148.194 N=2509046.334	920.100	920.054	1:100	R.C. PIPE DRAIN	ø200	
13	13	14	D12	9.38	E=263148.194 N=2509046.334	E=263157.336 N=2509044.220	920.054	920.000	1:175	R.C. PIPE DRAIN	ø300	
14	15	16	D13	7.8	E=263096.516 N=2509080.852	E=263092.381 N=2509087.470	920.150	920.130	1:400	EARTHEN TRAPEZOIDAL	300	
15	16	17	D14	52.11	E=263092.381 N=2509087.470	E=263136.573 N=2509115.078	920.130	920.000	1:400	EARTHEN TRAPEZOIDAL	300	
16	17	18	D15	48.3	E=263136.573 N=2509115.078	E=263162.164 N=2509074.117	920.000	919.879	1:400	EARTHEN TRAPEZOIDAL	300	
17	18	19	D16	27.8	E=263162.164 N=2509074.117	E=263176.895 N=2509050.538	919.877	919.830	1:600	EARTHEN TRAPEZOIDAL	300	
18	20	21	D17	42.0	E=263101.444 N=2509072.964	E=263123.695 N=2509037.348	920.150	920.066	1:500	EARTHEN TRAPEZOIDAL	300	
19	21	22	D18	17.0	E=263123.695 N=2509037.348	E=263132.703 N=2509022.931	920.066	920.032	1:500	R.C. PIPE DRAIN	ø300	
20	22	23	D19	37.18	E=263132.703 N=2509022.931	E=263164.238 N=2509042.630	920.032	919.958	1:500	EARTHEN TRAPEZOIDAL	300	
21	23	19	D20	14.93	E=263164.238 N=2509042.630	E=263176.895 N=2509050.538	919.958	919.838	1:125	EARTHEN TRAPEZOIDAL	300	
22	14	23	D21	7.08	E=263157.336 N=2509044.220	E=263164.238 N=2509042.630	920.000	919.960	1:175	EARTHEN TRAPEZOIDAL	300	
23	10	18	D22	6.12	E=263156.200 N=2509075.494	E=263162.164 N=2509074.117	919.897	919.877	1:300	EARTHEN TRAPEZOIDAL	300	



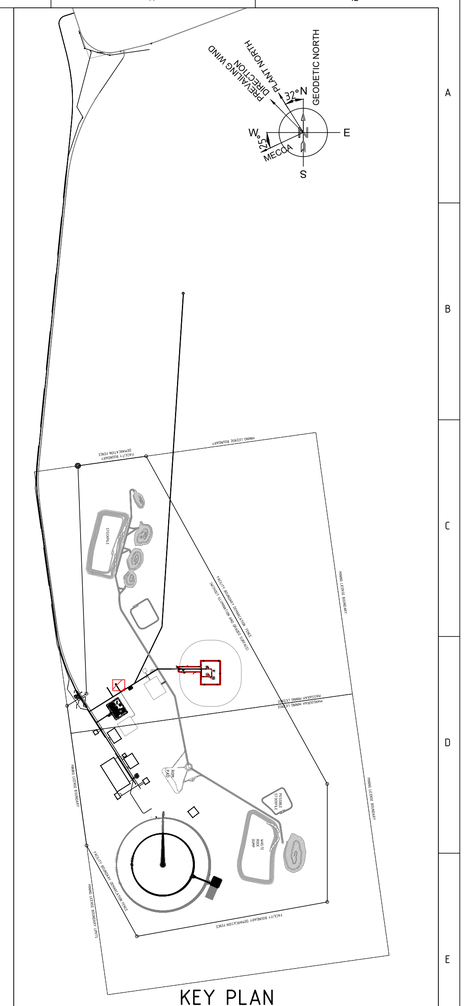
TYPICAL CROSS SECTION OF ROAD WITH EARTHEN DRAIN

SCALE= 1:50



TYPICAL G.A. DETAIL OF PIPE UNDER ROAD FOR STORM WATER DRAIN

SCALE= 1:50



KEY PLAN

LEGEND:

	FENCE
	NEW ROAD (L&T SCOPE)
FGL	FINISHED GROUND LEVEL
R.C.	REINFORCED CONCRETE
G.A.	GENERAL ARRANGEMENT
TYP	TYPICAL
P.C.C	PLAIN CEMENT CONCRETE
℄	CENTRE LINE
DIA	DIAMETER
C/C	CENTRE TO CENTRE
T/O	TOP OF
THK	THICK
LG.	LONG

NOTE:

1. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE SPECIFIED.
2. CO-ORDINATES AND LEVELS ARE IN METRES.
3. FINISHED GROUND LEVEL (F.G.L) = EL(+)920.450M.
4. GRADE OF REINFORCED CEMENT CONCRETE (R.C.C) SHALL BE M35.
5. REINFORCEMENT BARS SHALL CONFORM TO ASTM A615 GRADE 60 (fy=420 N/sq.mm.)
6. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE AS FOLLOWES:
   a) WALL = 50 mm.
   b) SLAB = 25 mm.
7. TYPICALLY D10-200 INDICATES DIA. OF BAR 10 mm AND SPACING 200 mm CENTRE TO CENTRE.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	16/02/2023	AS-BUILT	APEX	BJB	APEX	SJM	
		C01	20/02/2021	ISSUED FOR CONSTRUCTION	APEX	BJB	APEX	SJM	
		A02	18/01/2021	REVISED AS PER EMPLOYER'S COMMENT ON 29/12/2020	APEX	BJB	APEX	SJM	
MD-216-49EE-EG-CV-GAD-1584	ROAD LAYOUT & SECTIONS			& REVISED AS MARKED (1)					
MD-216-1000-EG-ME-PPD-1001	SITE MAP GENERAL OVERALL PROJECT LAYOUT	A01	25/11/2020	ISSUED FOR REVIEW	APEX	BJB	APEX	SJM	

RESERVER FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

This drawing is the property of *LARSEN & TOUBRO* and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Outotec

The content of these documents are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without prior written permission of Outotec is prohibited.

MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

This document and the information contained herein is Ma'aden proprietary know-how. This document or any part thereof should not be reproduced, stored in a retrieval system or transmitted by any means, electronic or otherwise, or divulged without prior written permission of Ma'aden.

LARSEN & TOUBRO DRAWING NUMBER: 019045_450_C_G_24281_001 — REVISION: 3

OUTOTEC DRAWING NUMBER: 900707__GAU01__AG90__MTL01__50035 — DOCUMENT ID: OU100706304 — REVISION: 3

DRAWN	APEX	20/11/2020
CHECKED	BJB	24/11/2020
ENGINEER	APEX	19/11/2020
APPROVED	SJM	25/11/2020
EMPLOYER		
SCALE	1:250	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
SEWAGE TREATMENT PLANT
LAYOUT & SECTIONS OF STORM WATER DRAIN

DRAWING NUMBER: MD-216-49EE-EG-CV-GAD-1585 — REV: D01